

**College of Engineering and Technology
Department of Computer Science**

Fundamentals of Database Systems

Chapter 2: Database System Architecture

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Outline

- ✓ Data models, Schema and Instances
- ✓ Database management system
- ✓ DBMS Architecture and data independence
- ✓ Database management system language and interfaces

Data models, Schema, and Instances

- **Data model** a collection of concepts that can be used to describe the **structure** of a database, the **operations** for manipulating these structures, and certain **constraints** that the database should obey.
- **Structure of the database means:** The **data types, relationships** and **Constraints** that applied to the data.
- Dynamic aspect or behavior of a database application.
- There are **two basic components** of the **database**. The **two components** are:
 1. *the definition of the Relation or the Table* and
 2. *the actual data stored in each table.*

Data Model Operations

- These operations are *used for specifying database retrievals and updates* by referring to the constructs of the data model.
- Operations on the data model may include :
 - ✓ ***basic model operations*** (e.g. generic insert, delete, update) and
 - ✓ ***user-defined operations*** (e.g. compute_student_gpa, update_inventory)

Data model structure and constraints

- ✓ Constructs are used to define the database structure.
- ✓ Constructs typically include elements as well as groups of elements (eg. Entity, record, table) and relationship among those groups.

Constraints:- specify some **restrictions** on valid data, these constraints must be enforced all times.

Categories of data model

- ✓ Conceptual/high-level
- ✓ Representational/implementation data models
- ✓ Physical/low-level data models

Conceptual data model:-

- ✓ Provide concepts that are close to the way any users perceive data.
- ✓ Also called entity-based or object based data model.
- Three basic **concepts of conceptual data model** are:
 - ✓ **Entity**: real world thing
 - ✓ **Attributes**: properties or characteristics of an entity
 - ✓ **Relationship**: association between two entities

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Example:

- **Customer** and **Product** are two entities. **Customer number** and **name** are attributes of the **Customer** entity
- **Product name** and **price** are attributes of product entity
- **Sale** is the relationship between the **customer** and **product**



■ Characteristics of conceptual data model:

- ✓ Offers Organization-wide coverage of the **business concepts**.
- ✓ This type of **Data Models** are designed and developed for a **business audience**.
- ✓ The **conceptual model** is developed **independently** of hardware specifications like **data storage capacity, location** or **software** specifications like **DBMS vendor and technology**.
- ✓ The focus is **to represent data as a user** will see it in the “**real world**.”

Representational/implementation data model:

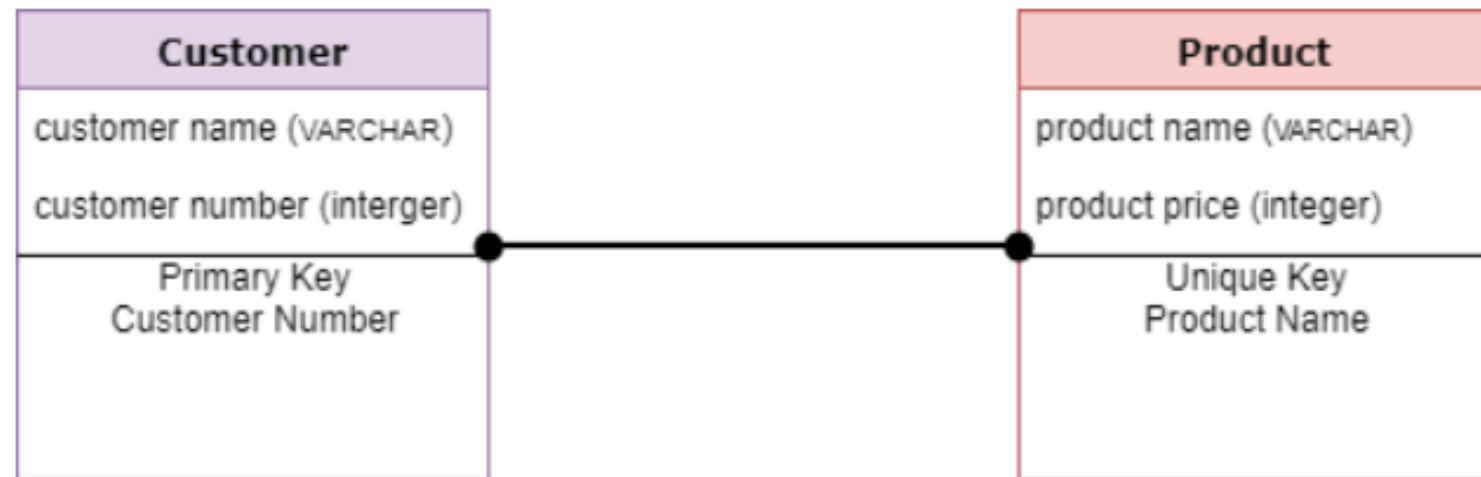
- ✓ This type of data model is used to represent only the **logical part** of the database and **does not** represent the **physical structure** of the databases.
- ✓ The representational data model allows us to focus primarily, on the **design part of the database**.
- ✓ A popular representational model is [Relational model](#).

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Physical data model:

- ✓ Provide concepts that describe **details** of how data is **stored in the computer**.
- ✓ It is representation of a data design as implemented, intended to be implemented in the **DBMS**.
- ✓ It has all the information of the format in which the file are present, and
- ✓ The structure of the databases, presence of external data structures and their relationships.

Example:



Characteristics of physical data model:

- ✓ The physical data model describes data need for a **single project or application** though it maybe integrated with other **physical data models** based on **project scope**.
- ✓ Data Model contains relationships between tables that which addresses **cardinality** and **nullability** of the relationships.
- ✓ Developed for a specific version of a **DBMS, location, data storage or technology** to be used in the project.
- ✓ Columns should have exact **datatypes, lengths** assigned and **default** values.
- ✓ **Primary and Foreign keys, views, indexes, access profiles, and authorizations, etc. are defined.**

Conceptual ERD

- Tables (entities) and field names (attributes)
- Relationships

Logical ERD I

- Tables (entities) and field names (attributes)
- Relationships
- **Datatypes**

Physical ERD

- Tables (entities) and field names (attributes)
- Relationships
- Datatypes
- **Field sizes** (lengths)
- **Key fields**

Schema and Instances

- **Database schema:-** description of database is called **database schema**.
 - ✓ It describes **how data** is structured.
 - ✓ Specifies **name of relation** and **collection** of attributes.
 - ✓ Specified during **database design** and should **not be changed** unless **during maintenance**.
- **Schema construct:** Each object in the schema such as **student**, **course** are *schema construct*.

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- Example of database schema

STUDENT

Name	Student_number	Class	Major
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COURSE

Course_name	Course_number	Credit_hours	Department
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PREREQUISITE

Course_number	Prerequisite_number
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Database **instances/snapshot/extension**:

- Also known as **current set** of occurrences.
- The data in the database at a **particular moment** in time is called **instance**.
- Changes **rapidly**

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Example of
database instances:

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

Database Management System

- ***Database Management System (DBMS)*** is a Software package used for providing EFFICIENT, CONVENIENT and SAFE MULTI-USER storage of and access to MASSIVE amounts of PERSISTENT data.
- A *DBMS* also provides a systematic method for:
 - ✓ creating,
 - ✓ updating,
 - ✓ storing,
 - ✓ retrieving data in a database.

- Having this in mind, **a full scale DBMS should at least have the following services** to provide to the user.
 - ✓ *Data storage, retrieval and update* in the database
 - ✓ A user accessible *catalogue* .
 - ✓ *Concurrency Control Services*: access and update on the database by **different users simultaneously** should be implemented correctly.
 - ✓ *Recovery Services*: a mechanism for recovering the database after a **failure** must be available.

- ***Authorization Services*** (Security): must support the implementation of access and authorization service to database administrator and users.
- ***Integrity Services***:
 - ✓ Services to promote *data independency* between the **data** and the **application**
- ***Utility services***: sets of utility service facilities like
 - ✓ *Importing data*
 - ✓ *Statistical analysis support*
 - ✓ *Index reorganization*
 - ✓ *Garbage collection*

DBMS architecture and data independence

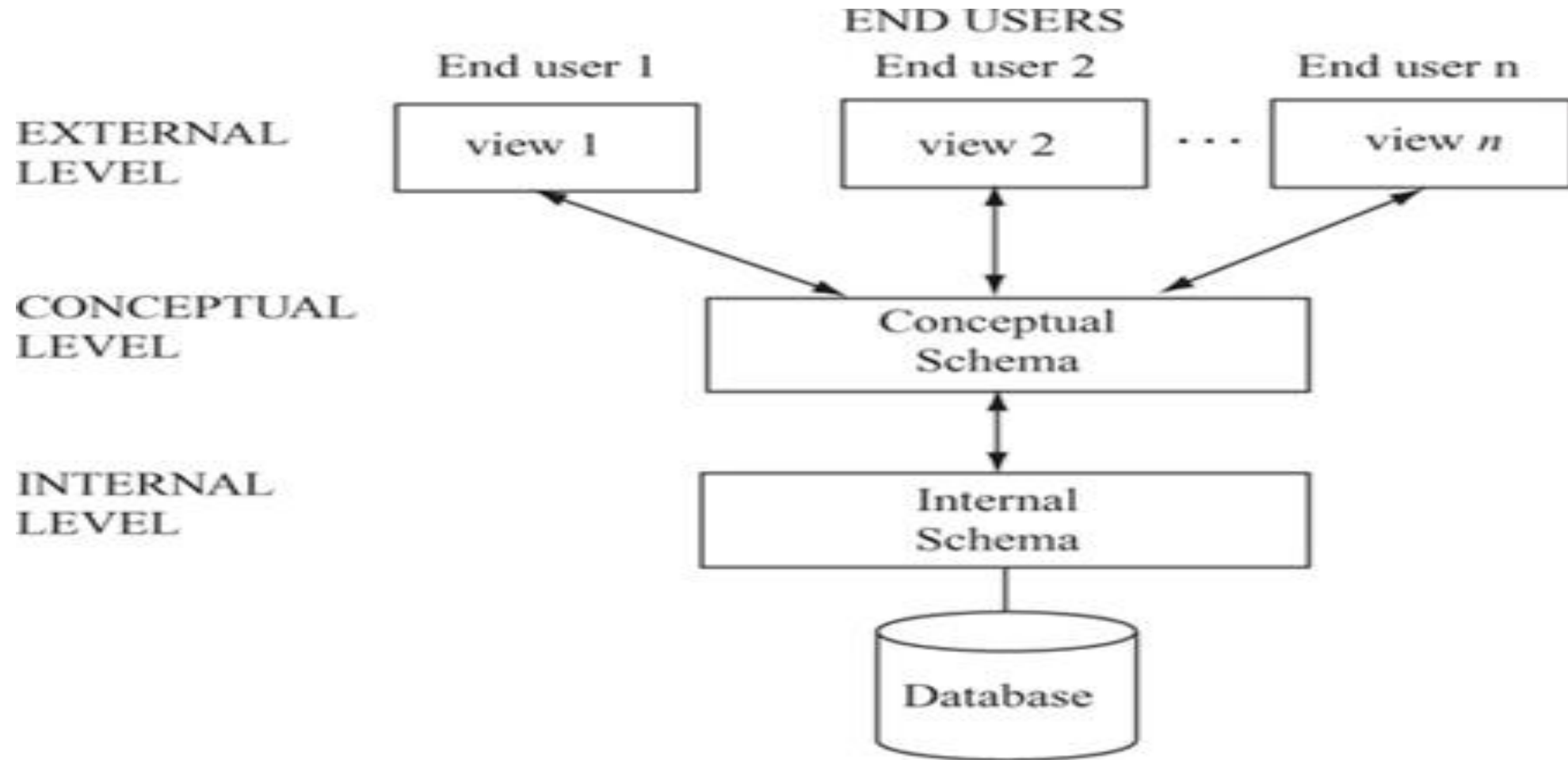
- A **DBMS architecture** allows dividing the database system into individual components that can be independently **modified, changed, replaced, and altered**.
- It describes how data in the **database is viewed** by the **users**.
- It **is not** concerned with how the data is **handled** and **processed** by the **DBMS**.

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- This enables the users to manipulate the data without worrying about where it is located or how it is actually stored.
- In this architecture, the overall database description can be defined at three levels, namely,
 - ✓ *internal*,
 - ✓ *conceptual*, and
 - ✓ *external* levels, and thus, named three-level DBMS architecture.
- This architecture is proposed by ANSI/SPARC (American National Standards Institute/Standards Planning and Requirements Committee) and hence, is also known as [ANSI/SPARC architecture](#).

DBMS Architecture

- The three schema-architecture



- Internal level:

- ✓ Describes physical storage structure of the database and also known as physical level.
- ✓ At this level, various aspects are considered to achieve optimal runtime performance and storage space utilization. These aspects include:
 - Storage space allocation techniques for data and indexes, access paths such as indexes, data compression and encryption techniques, and record placement.

Conceptual level:

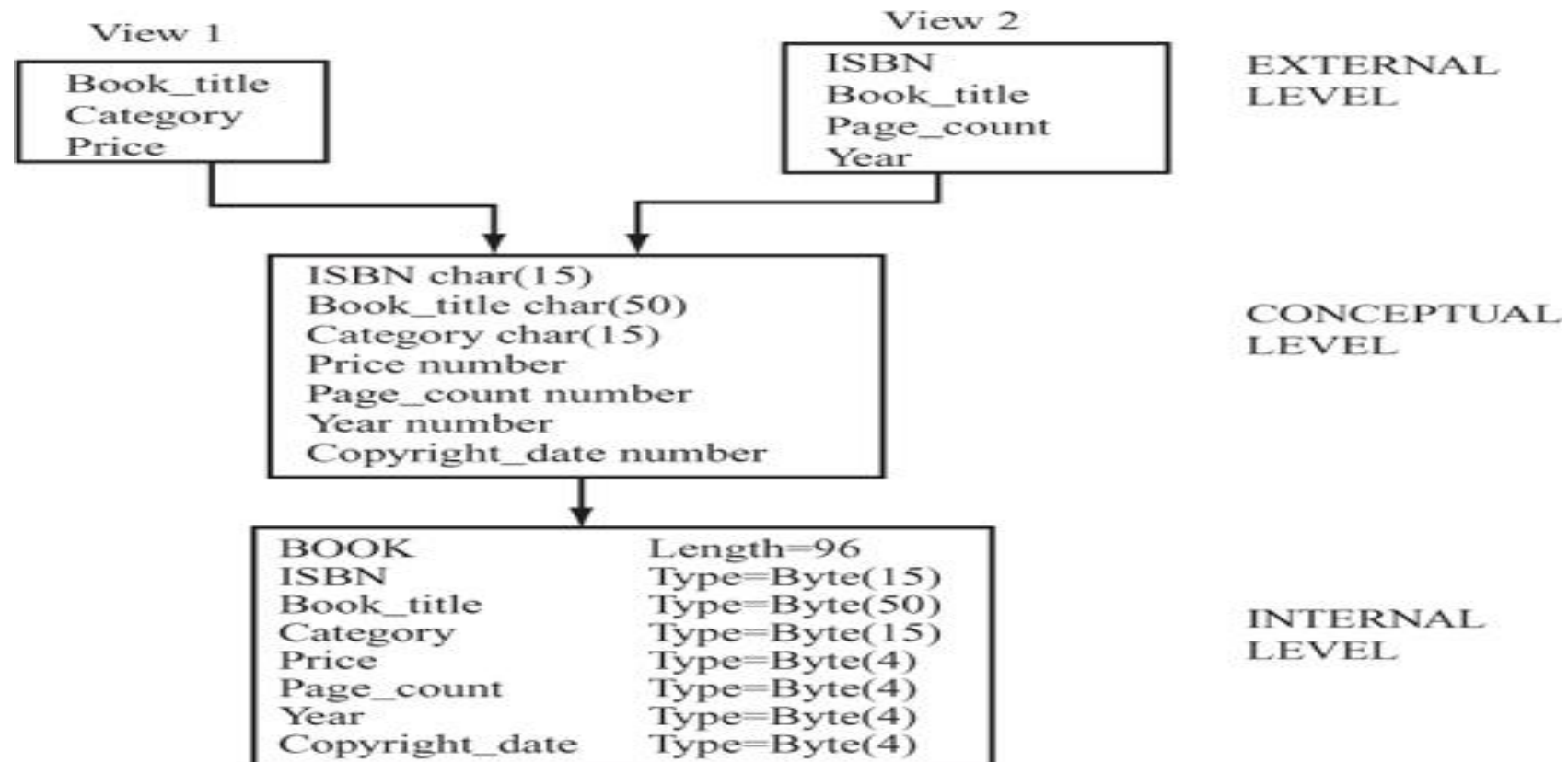
- Describes structure of the whole database for the whole users.
- It describes what data is stored in the database and relationship among the data.

External level:

- Describes part of database of interest to a particular user group.
- It is users view of the database.
- Different users have their own customized view of the database.

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- To understand the three-schema architecture, consider the three levels of the **BOOK** file in *Online Book* database as shown below:



Data independence

- One of the main advantage of the **three-schema architecture** is it provides data independence.
- **Data independence** is the ability to change the schema at one level of the database system without having to change the schema at the other level.

DBMS languages and Interfaces

- ❖ Because a database supports a number of user groups, as mentioned previously, the DBMS must have languages and interfaces that support each user group.
- ❖ DDL(**data definition language**), used by the **DBA** and database designers to define the conceptual and internal schemas.
 - ✓ Commands for setting up schema or intention of the database.
 - ✓ These commands are used to setup a database, create, delete, and alter table with the facility of handling constraints.

DBMS languages and Interfaces

- ✓ **SDL**(storage definition language):is used to specify the internal schema.
- ✓ **VDL**(view definition language):is used to specify the user views and their mappings to the conceptual schema.
- ✓ **DML**(data manipulation language):Manipulations include retrieval, insertion, deletion and modification.

DBMS Interfaces: The following are some DBMS interfaces,

- *Menu-Based Interfaces for Web Clients or Browsing*

- ✓ Present users with list of options (menus)
- ✓ Lead user through formulation of request
- ✓ Query is composed of selection options from menu displayed by system.

- *Forms-Based Interfaces*

- ✓ Displays a form to each user.
- ✓ User can fill out form to insert new data or fill out only certain entries.
- ✓ Designed and programmed for naïve users as interfaces to canned transactions.

❖ *Graphical User Interfaces*

- ✓ Displays a schema to the user in diagram form. The user can specify a query by manipulating the diagram. GUIs use both forms and menus.

❖ *Natural Language Interfaces*

- ✓ Accept requests in written English, or other and attempt to understand them.
- ✓ Interface has its own schema, and a dictionary of important words. Uses the schema and dictionary to interpret a natural language request.

❖ *Interfaces for Parametric Users*

- ✓ Parametric users have small set of operations they perform.
- ✓ Analysts and programmers design and implement a special interface for each class of naïve users.
- ✓ Often a small set of commands included to minimize the number of keystrokes required. (I.e. function keys)

❖ *Interfaces for the DBA*

- ✓ Systems contain privileged commands only for DBA staff.
- ✓ Include commands for creating accounts, setting parameters, authorizing accounts, changing the schema, reorganizing the storage structures etc.
- ✓ Database System Environment

End of Chapter 2

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Thank You!

Individual Assignment(5%)

#Do all the questions below accordingly

- 1). Describe clearly the Centralized and Client-server DBMS Architecture?
- 2). Describe the following DBMS tier Architecture
 - a) 1-tier DBMS architecture
 - b) 2-tier DBMS architecture
 - c) 3-tier DBMS architecture
- 3). To design and use a database, there will be the interaction or integration of database system components. Therefore explain clearly and briefly the DBMS components and how they interacting each other?